Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H1 BIOLOGY

8875/02

Paper 2Structured and Free-Response Questions

15 September 2014

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section B Structured Questions

Answer **all** questions.

Write your answers on space provided in the Question Paper.

Section C Free-Response Questions

Answer **one** question. Your answer to Section C must be in continuous prose, where appropriate. Write your answers on the writing paper provided.

Submit your answers to Sections B and Section C separately.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [ ].

For Examiner's Use	
Section A [30]	
Section B [40]	
1	**/ 5**
2	**/5**
3	**/ 10**
4	**/ 5**
5	**/ 5**
6	**/ 10**
Section C [20]	
1 / 2	
Total [90]	

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For Examiner's use

Section B: Structured Questions (40 marks)
Answer **all** questions in this section.

Question 1

(a) **Fig1** shows an electron micrograph of a cell from the lung that is specialised to take up material by endocytosis. Darkly staining vesicles containing inorganic aluminiumsilicate crystals are labelled in **Fig1**. These crystals are only ever seen in such cells in the lungs of cigarette smokers. Aluminium silicate is known to be present in tobacco smoke.

Fig1

(i) Explain how endocytosis allows foreign material such as aluminium silicate to be taken up by cells. [2]

(ii) Describe how foreign material is normally dealt with once in these vesicles. [3]

Total: [5]

Question 2

For Examiner's use

Fig 2shows some cells undergoing mitosis. Each of the cells **A**, **B**, **C** and **D** is in a different stage of mitosis.

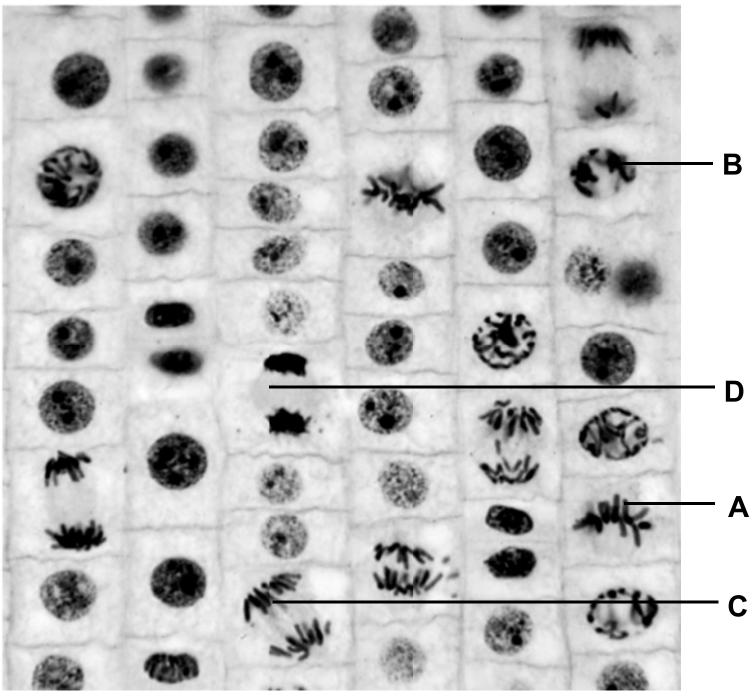


Fig 2

(a) Using the letters provided, write the correct order of the stages in mitosis. [1]

(b) Describe what happens to the nuclear envelope during mitosis. [2]

(c) Paclitaxel is a drug used in the treatment of cancer. It stabilizes the microtubule polymer and protects it from disassemblyduring nuclear division. Explain how treatment with paclitaxel will affect the behavior of chromosomes during nuclear division and how it serves as an effective treatment. [2]

Total: [5]

For Examiner's use

Question 3

(a) Protein synthesis in a eukaryotic cell is a multi-step process. **Fig 3.1** below shows a step during protein synthesis.

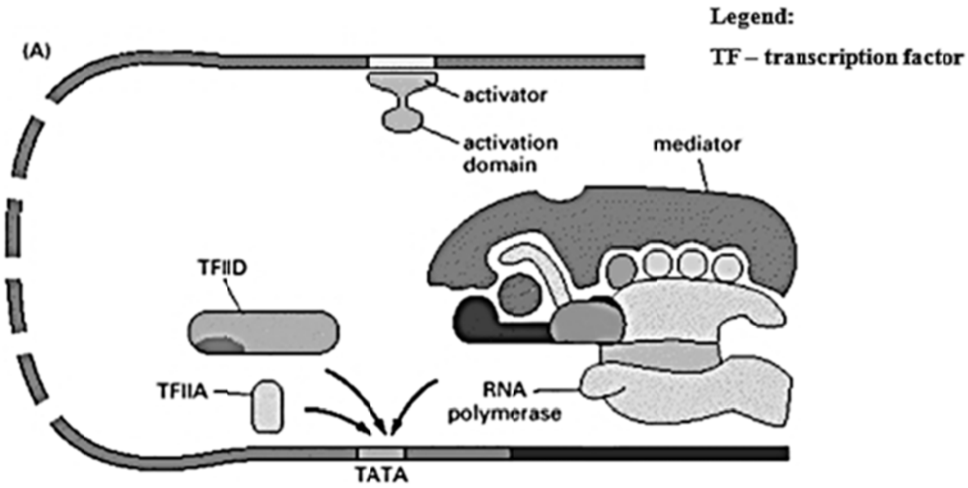


Fig 3.1

(i) State precisely where this process can be found. [1]

____

(ii) With reference to **Fig 3.1**, describe how protein synthesis will be affected when there is a mutation to the TATA region. [3]

____

____

____

____

____

____

For Examiner's use

(b) **Fig 3.2** shows the binding of RNA polymerase to DNA and the sequences of DNA which the RNA polymerase will transcribe.

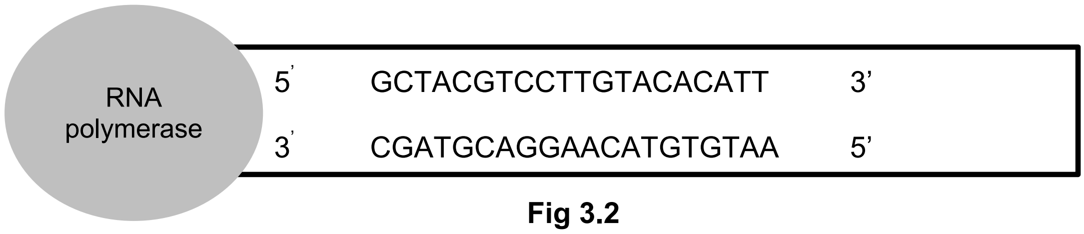


Fig 3.2

(i) Write down the mRNA sequence transcribed. [1]

(ii) Explain how you arrive at your answer in (i). [2]

(c) Describethree ways in which transcription differs from translation in protein synthesis. [3]

Total: [10]

For Examiner's use

Question 4

(a) Distinguish between the terms *dominant* and *recessive*. [1]

(b) Tay-Sachs disease is a recessive hereditary abnormality causing death within the first few years of life only when homozygous (aa). The dominant condition at this locus produces a normal phenotype (A-). Abnormally shortened fingers (brachyphalangy) are thought to be due to a genotype for a lethal gene (BB^L), the homozygote (BB) being normal, and the other homozygote (B^LB^L) being lethal.

What are the phenotypic expectations among teenage children from parents who are both brachyphalangic and heterozygous for Tay-Sachs disease.

Construct a genetic diagram to illustrate your answer. You can use the space below to construct the genetic diagram. [4]

Total: [5]

For Examiner's use

Question 5

The waters of South-Eastern Australia are home to a group of sea slugs found nowhere else. Ten distasteful species of the genus *Chromodoris*all share a pattern of bright red spots on a white background.

This is an example of mimicry (typified by a situation where a harmless species has evolved to imitate the warning signals of a harmful species) against their predator (fish).

(a) Suggest why one warning pattern shared by ten species gives a greater selective advantage to the slugs than if each had its own distinctive warning pattern. [1]

(b) Describe how a South-Eastern Australian sea slug population that was originally white might have developed the red-spotted pattern over a period of time. [4]

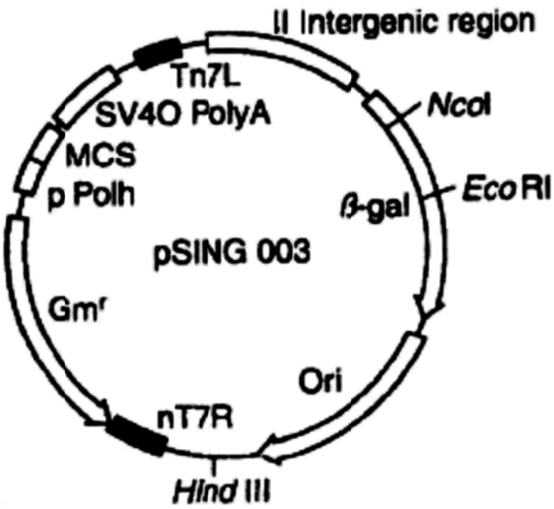
Total: [5]

For Examiner's use

Question 6

(a) Scientists have recently produced genetically-engineered clover containing a protein high in sulphur-containing amino acids. A DNA fragment was prepared for insertion into the clover.

Fig 6.1 shows a vector map of the plasmid vector pSING003. The plasmid is used to clone the DNA fragment, which will be inserted at the *Eco*RI site. The plasmid is subsequently introduced into *E. coli* via transformation. After transformation, the *E. coli* cells are then grown on an agar plate.



Gm^r – gene coding for resistance to the antibiotic gentamycin

β-gal – gene coding for the enzyme β-galactosidase

Fig 6.1

(i) Describe two properties of plasmids that allow them to be used as DNA cloning vectors. [2]

(ii) In addition to the nutrients required for growth, name two other substances that are added to the agar plate to allow selection of cells successfully transformed with the recombinant plasmid. [1]

1.

2.

(b) **Fig 6.2** shows the results of plating the transformed cells onto an agar plate with the appropriate substances.

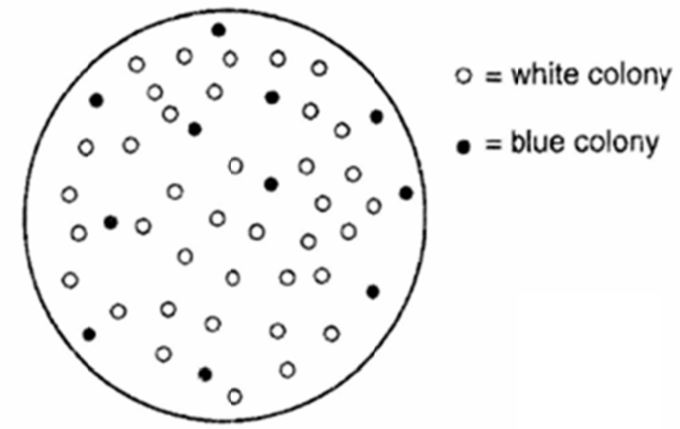


Fig 6.2

(i) Explain why some colonies appeared white while others appeared blue. [4]

(ii) Using **Fig 6.2**, calculate the percentage of cells into which the recombinant plasmid had been successfully introduced. Show your working in the space provided below. [1]

(iii) Suggest why replica plating was not necessary in this experiment. [2]

Total: [10]

Section C: Free-Response Question(20 marks)

Answer only **one** question.
Write your answers on the writing paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
A **NIL RETURN** is required.

Question 1

(a) Describe the effects of a mutation of one named gene. [10]

(b) **(i)** Using appropriate examples, explain the normal functions of stem cells in living organisms. [6]

(ii) Discuss the importance of cell cycle checkpoints in the self-renewal of stem cells. [4]

OR

Question 2

(a) Distinguish between the processes of Krebs Cycle and Calvin Cycle. [7]

(b) Describe the role of oxidative phosphorylation. [7]

(c) Discuss the effects of varying light intensity on photosynthesis. [6]

Total: [20]

END OF PAPER

DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2014
YEAR SIX
H1 BIOLOGY (8875)

Suggested Answers

1(a)(i)
Phagocytosis
uptake of particulate material involving pseudopodia and engulfing of material
Formation of vesicles around the material involves the fusion of cell surface membrane as it pinched off

(ii)
primary lysosomes containing hydrolytic enzymes (e.g. lipases, proteases, carbohydrases, nucleases)
fuse with vacuoles containing engulfed material / phagosome forming secondary lysosome
subsequent breakdown by hydrolysis of that material to soluble products

2(a)
B, A, C, D

(b)
During prophase, nuclear envelope starts to fragment and disintegrate
During telophase, the nuclear envelope re-forms around the chromosomes at each pole

(c)
Paclitaxel prevents the kinetochore microtubules from shortening during anaphase
as a result the sister chromatids cannot be pulled to the opposite poles of the cell and mitosis cannot proceed
This prevents proliferation (rapid cell division) / multiplication of cells through mitosis of cancerous cells

3(a)(i)
Nucleus

(ii)
TFIID/ TFIIA (general transcription factors) unable to recognise and bind promoter region
RNA polymerase unable to recognize and bind to promoter, transcription initiation complex cannot be formed
Transcription cannot occur and protein synthesis will be decreased

(b)(i)
5’ GCUACGUCCUUGUACACAUU 3’

(ii)
RNA polymerase reads the DNA template strand in a 3’ to 5’ direction
Newly synthesised RNA is added based on complementary base pairing, A:U, T:A and C:G

(c)

	Transcription	Translation
Location	nucleus	cytoplasm
template	DNA template strand	mRNA
products	mRNA, rRNA, tRNA;	Polypeptide / protein;
bonds formed between	phosphodiester bond is formed between monomers;	peptide bond is formed between;
monomers	Free ribonucleotide triphosphate / ribonucleotides;	amino acids;
Enzyme	RNA polymerase;	Peptidyl transferase;

4(a)
Dominant trait is expressed when at least 1 dominant allele is present in the genotype while recessive trait is expressed only in the absence of dominant allele in genotype / organism has two copies of the recessive allele;

(b)

Parental phenotypes : brachyphalangic and heterozygous for Tay-Sachs disease x brachyphalangic and heterozygous for Tay-Sachs disease

Parental genotypes : $AaBB^L$ x $AaBB^L$ 1

Gametes produced by parents : ¼ AB ¼ AB^L ¼ aB ¼ aB^L ; ¼ AB ¼ AB^L ¼ aB ¼ aB^L 1

Punnett square:

	¼ AB	¼ AB^L	¼ aB	¼ aB^L
¼ AB	1/16 AABB	1/16 $AABB^L$	1/16 AaBB	1/16 $AaBB^L$
¼ AB^L	1/16 $AABB^L$	1/16 AAB^LB^L	1/16 $AaBB^L$	1/16 AaB^LB^L
¼ aB	1/16 AaBB	1/16 $AaBB^L$	1/16 aaBB	1/16 $aaBB^L$
¼ aB^L	1/16 $AaBB^L$	1/16 AaB^LB^L	1/16 $aaBB^L$	1/16 aaB^LB^L

Genotypes of offspring:	1 AABB 2 AaBB	2 $AABB^L$ 4 $AaBB^L$	1 aaBB	2 $aaBB^L$	1 aaB^LB^L 1 AAB^LB^L 2 AaB^LB^L 1
Phenotypes of offspring:	Normal: No Tay-Sachs disease and not brachyphalangic	No Tay-Sachs disease but brachyphalangic	With Tay-Sachs disease but not brachyphalangic	With Tay-Sachs disease and brachyphalangic	Death by lethal gene 1
Phenotypic ratio of offspring:	3 :	6	Death in first few years	Death in first few years	

5(a)
ref to fish learning where shared pattern minimises fish attacks / AW

(b)
natural selection where predation (by fish) is the selective pressure
random / spontaneous / chance, mutation in gene is source for variation which produces phenotype e.g. for red spots / pigment (protein)
those with red spots escapes predation, reproduce and pass on (red-spot), allele / mutated gene
allele increases in frequency in population and further mutations improving pattern selected for

6(a)(i)
Possess own origin of replication thus can replicate independently of main chromosome;
Possess restriction sites which can be cut by restriction enzymes for insertion of gene-of-interest

(ii)
1. X-gal
2. Antibiotic gentamycin

(b)(i)
Intact β-galactosidase gene codes for enzyme which hydrolyzes X-gal (giving rise to blue colour)
Bacteria colonies with reannealed plasmids will appear blue
Bacteria colonies with recombinant plasmids will appear white
Due to insertional inactivation which disrupts β-galactosidase gene

(ii)
Percentage of cells = 38/49 X 100 = 77.6% (correct to 3 sig. fig.)

(iii)
Replica plating is only needed if the bacteria colonies which possess recombinant or reannealed plasmids are cultured on the same medium and cannot be differentiated
Bacteria with recombinant plasmids can be differentiated visually by their colours

Essay

Marking Point	Answer
1	Single base / nucleotide substitution within the gene;
2	thymine substitutes with adenine within the gene coding for ß-globin chain / sequence in gene is changed from CTT to CAT;
3	mRNA codon is changed from GAA to GUA;
4	Results in a change of amino acid, glutamic acid is replaced by valine;
5	Primary structure of ß-globin chain is altered, leading to change in folding and coiling of the polypeptide;
6	Glutamic acid carries a negative charge & is polar / hydrophilic, while valine is non polar and hydrophobic;
7	substitution creates a hydrophobic spot on the outside of the protein structure that sticks to the hydrophobic region of an adjacent hemoglobin molecule's beta chain;
8	HbS molecule aggregates/ crystalises under low oxygen conditions, so is inefficient in carrying oxygen;
9	And causing red blood cells to change from circular biconcave shape to sickle shape;
10	Sickle shaped red blood cells can obstruct blood vessels and haemolyse readily, leading to phenotype of anaemia;
11	Individuals with 2 copies of the Hb^S alleles / genotype Hb^SHb^S display severe symptoms of sickle cell anemia; OR Carriers with genotype Hb^SHb^A only have symptoms if they are deprived of oxygen (for example, while climbing a mountain) as the normal gene is able to produce 50% of the hemoglobin in the body;

(b)(i)

Marking Point	Answer
12	Zygotic stem cells are formed when an sperm fuses with an oocyte;
13	The totipotent zygotic stem cells can differentiate and give rise to any cell type;
14	They are necessary for the formation and development of a complete fetus;
15	Embryonic stem cells are found in the inner cell mass of the blastocyst;.
16	The pluripotent embryonic stem cells can differentiate and give rise to any cell type except the embryonic support structures;
17	They are necessary for the formation of a complete embryo;
18	Adult stem cells are important in growth and regeneration of the body;
19	E.g: Blood stem cells are found in the bone marrow;

20	The multipotent blood stem cells can differentiate and give rise to platelets, white blood cells and red blood cells;
21	These new cells replace the cells that are lost through normal wear and tear, injury or disease;

(ii)

Marking Point	Answer
22	Stem cells divide only when appropriate growth signal is present;
23	Cell cycle checkpoints control the timing, number and rate of cell division;
24	Prevent uncontrolled division;
25	G1/S checkpoint checks for adequate cell size, replication of organelles and integrity of DNA;
26	G2/M checkpoint checks for accuracy in DNA replication;
27	Spindle checkpoint checks for the attachment of spindle fibres to centromeres of chromosomes;
28	Ensure that daughter cells will be genetically identical to parent cell;
29	Daughter stem cell will have the same characteristics as the parent stem cell;

2(a)

Marking Point		Krebs cycle	Calvin cycle
30	Location	Mitochondrial matrix	Chloroplast stroma
31	Substrate	Acetyl-CoA and oxaloacetate combines to form citrate	CO_2 and Ribulose bisphosphate (RuBP)
32	Products	Each glucose molecule gives rise to: 6 NADH 2 $FADH_2$ 2 ATP 4 CO_2	For every 3 molecules of CO_2 that enter the cycle, one triose phosphate / G3P is made
33	Regenerated / Starting material	Oxaloacetate is the starting material that is eventually regenerated	Ribulose bisphosphate (RuBP) is the starting material that is eventually regenerated
34	ATP	Produced via susbstrate level phosphorylation	Used in reduction of glycerate-3-phosphate where energy Is required through hydrolysis of ATP
35	Electron carriers / donors	Use NAD^+ and FAD for the oxidation of the intermediates of the cycle by serving as electron acceptors	Uses NADPH / reduced $NADP^+$ to reduce glycerate-3-phosphate to triose phosphate by serving as electron donors
36	Overall	Catabolic	Anabolic
37	Role of CO_2	CO_2 is released as a result of decarboxylation reactions	Required for carbon fixation. CO_2 is used to

			convert Ribulose bisphosphate (RuBP) to form an unstable 6C compound that breaks down to form glycerate-3-phosphate
38	Role of O_2	Occurs only when O_2 is present	Does not require O_2

(b)

Marking Point	Answer
39	Oxidative phosphorylation is the synthesis of ATP in the presence of oxygen and takes place within the inner mitochondrial membrane;
40	Reduced NAD and reduced FAD provide high energy electrons for the synthesis of ATP;
41	Electrons are passed along the Electron Transport Chain (ETC) from one electron carrier to the next, each with an energy level lower than the one preceding it;
42	As the electrons are passed from one electron carrier to the next, some of their energy is used to transport protons from matrix of the mitochondrion into the intermembrane space;
43	This produces a high concentration of H^+ in the intermembrane space, setting up an electrochemical proton gradient / proton motive force;
44	Stalk particles containing ATP synthases are embedded on inner mitochondria membrane;
45	H^+ diffuse through them, down their electrochemical gradients, back into the matrix;
46	This provides enough energy to synthesise ATP by the phosphorylation of ADP with inorganic phosphate;
47	Oxygen functions as the final proton and electron acceptor to form water, catalyzed by cytochrome oxidase;
48	NAD^+ and FAD are regenerated at the end of oxidative phosphorylation;

(c)

Marking Point	Answer
49	Draw curve / graph of effects of varying light intensity on rate of photosynthesis + label light saturation point)
50	Light intensity directly affects light-dependent stage of photosynthesis. When light energy is absorbed by chlorophyll, electrons are excited to a higher energy level to be passed down the electron transport chain.
51	The process of photophosphorylation is responsible for both the photolysis of water to produce O_2 and the synthesis of ATP and NADPH to be used for Calvin Cycle where CO_2 is taken up.
51a	The rate of photosynthesis is often measured by the amount of CO_2 absorbed or O_2 evolved by the plant.
52	At lower light intensities, light intensity is the limiting factor where rate of photosynthesis increases proportionally with increasing light intensity
53	With increasing light intensity, compensation point is reached where the amount of carbon dioxide produced in respiration exactly balances that being used in photosynthesis
54	Further increase in light intensity results in the light saturation point being reached where light intensity will have no effect on the rate of photosynthesis
55	Light intensity is no longer a limiting factor, other environmental factors like CO_2 conc or temperature may be limiting now.